

Xu Wang Yin

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Education

08/2016 - 12/2022	Ph.D, Computer Engineering, University of Virginia, United States Ph.D thesis: Generative Modeling With Adversarial Training Supervisor: Prof. Gustavo Rohde Research focus: adversarial machine learning, energy-based generative models, multimodal models, optimal transport in signal processing
2011 - 2014	M.Eng, Computer Engineering, University of Science and Technology Beijing, China Master thesis: Robust Text Detection in Natural Scene Images Supervisor: Prof. Xu-Cheng Yin
2007 - 2011	B.Eng, Computer Science and Engineering, University of Science and Technology Beijing, China

Employment

04/2025 - Present	Independent Researcher (San Francisco/Qingdao) - Developed a novel adversarial training framework for energy-based models enabling joint image generation and classification in a single model - First to scale energy-based models to ImageNet 256×256 with state-of-the-art sample quality and 5-29× faster inference than diffusion models - Single model supports generation, classification, OOD detection, and counterfactual explanations - Paper accepted to ICLR 2026 (arXiv:2510.13872)
01/2023 - 04/2025	Research Engineer, Center for AI Safety, San Francisco, United States - Trained harmful textual content detection model with LoRA + SFT + linear probe - Developed model evaluation library with transformers, litellm and vllm [Utility Engineering] - Evaluating base and chat LLMs using HELM, MACHIAVELLI, lm-evaluation-harness on capability benchmarks and safety benchmarks [Safetywashing] - Adversarial training of LLMs against GCG jailbreak prompts [HarmBench] - Implemented adversarial attacks against open weights multimodal LLMs [HarmBench] - Model steering using control vector computed from embedding features of LLMs [Representation Engineering] - Implemented latent variable-based adversarial attacks, integrated state-of-the-art vision models (CLIP, DINO, Vision Transformer, ConvNeXt) into codebase and conducted extensive model evaluation [Unforeseen Adversaries]
2014 - 2015	Computer Vision Engineer, Yuanli Education Technology, Beijing, China Developed neural network models for detecting and recognizing text in images

Research Interests

Energy-based generative models

- Scalable training of energy-based models via adversarial training
- Text-to-image generation with energy-based models
- Hybrid discriminative-generative models

Machine learning robustness and security

- Training adversarially robust image models and LLMs
- Safety evaluation/red-teaming/jailbreaking LLMs
- Detecting adversarial examples

Selected Publications

- Scalable Energy-Based Models via Adversarial Training: Unifying Discrimination and Generation
Xu Wang Yin, Claire Zhang, Julie Steele, Nir Shavit, Tony T. Wang
International Conference on Learning Representations (**ICLR 2026**) [[arxiv](#)]
- Learning Globally Optimized Language Structure via Adversarial Training
Xu Wang Yin
Working paper, September, 2023 [[arxiv](#)]
- Learning Energy-Based Models With Adversarial Training
Xu Wang Yin, Shiyong Li, Gustavo K. Rohde,
European Conference on Computer Vision (**ECCV 2022**) [[code](#)] [[arxiv](#)] [[poster](#)] [[video](#)]
- GAT: Generative Adversarial Training for Adversarial Example Detection and Robust Classification
Xu Wang Yin, Soheil Kolouri, Gustavo K. Rohde
International Conference on Learning Representations (**ICLR 2020**) [[code](#)][[arxiv](#)]
- OBJ2TEXT: Generating Visually Descriptive Language from Object Layouts. [[code](#)][[Link](#)]
Xu Wang Yin and Vicente Ordonez.
Empirical Methods in Natural Language Processing (**EMNLP 2017**, oral presentation)
- Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs [[code](#)][[arxiv](#)]
Mantas Mazeika, **Xu Wang Yin**, Rishub Tamirisa, Jaehyuk Lim, Bruce W. Lee, Richard Ren, Long Phan, Norman Mu, Adam Khoja, Oliver Zhang, Dan Hendrycks
NeurIPS 2025
- RenderAttack: Hundreds of Adversarial Attacks Through Differentiable Texture Generation
Dron Hazra, Alex Bie, Mantas Mazeika, **Xu Wang Yin**, Andy Zou, Dan Hendrycks, Maximilian Kaufmann
NeurIPS 2024 Workshop on New Frontiers in Adversarial Machine Learning
- HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal [[arxiv](#)]
[[website](#)]
Mantas Mazeika, Long Phan, **Xu Wang Yin**, Andy Zou, Zifan Wang, Norman Mu, Ellie Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, Dan Hendrycks

International Conference on Machine Learning (ICML 2024)

- End-to-End Signal Classification in Signed Cumulative Distribution Transform Space [[arxiv](#)]
AMH Rubaiyat, Shiyong Li, **Xu Wang Yin**, Mohammad Shifat E Rabbi, Yan Zhuang, Gustavo K. Rohde
IEEE Transactions on Pattern Analysis and Machine Intelligence (**PAMI 2023**)
- A Digital Camera-based Eye Movement Assessment Method for NeuroEye Examination [[link](#)]
Mohamed Abul Hassan, **Xu Wang Yin***, Yan Zhuang*, Chad M Aldridge, Timothy McMurtry,
Andrew M Southerland, Gustavo K Rohde
IEEE Journal of Biomedical and Health Informatics, 2023
- Radon Cumulative Distribution Transform Subspace Modeling for Image Classification
M Shifat-E-Rabbi, **Xu Wang Yin**, Abu Hasnat Mohammad Rubaiyat, Shiyong Li, Soheil Kolouri, Akram Aldroubi, Jonathan M. Nichols, Gustavo K. Rohde
Journal of Mathematical Imaging and Vision, 2021 [[code](#)][[Link](#)]
- Neural Networks, Hypersurfaces, and Radon Transforms
Soheil Kolouri, **Xu Wang Yin**, Gustavo K. Rohde
IEEE Signal Processing Magazine, 2020 [[code](#)][[IEEEExplore link](#)][[arxiv](#)]
- Robust Text Detection in Natural Scene Images.
Xu-Cheng Yin, **Xu Wang Yin**, Kaizhu Huang, Hong-Wei Hao.
IEEE Transactions on Pattern Analysis and Machine Intelligence (**PAMI 2014**) [[link](#)]
- Effective text localization in natural scene images with MSER, geometry-based grouping and AdaBoost.
Xu Wang Yin, Xu-Cheng Yin, Hong-Wei Hao, Khalid Iqbal.
International Conference on Pattern Recognition (**ICPR 2012**) [[IEEEExplore link](#)]
- Representation Engineering: A Top-Down Approach to AI Transparency [[arxiv](#)]
Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, **Xu Wang Yin**, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, Dan Hendrycks
- Testing Robustness Against Unforeseen Adversaries [[arxiv](#)] [[github](#)]
Max Kaufmann, Daniel Kang, Yi Sun, Steven Basart, **Xu Wang Yin**, Mantas Mazeika, Akul Arora, Adam Dziedzic, Franziska Boenisch, Tom Brown, Jacob Steinhardt, Dan Hendrycks

Awards

- [Top Reviewer of NeurIPS 2022](#)
- International Conference on Document Analysis and Recognition (ICDAR2013) robust reading competition [winner](#) (out of 15 research teams from 7 countries)
- Outstanding Graduate of Beijing, Beijing Municipal Commission of Education, Jan 2014.

Professional Service

ACL-IJCNLP 2021, EMNLP 2021, ACL 2022, ICLR 2022, ECCV 2022, NeurIPS 2022, NeurIPS 2024, ACL 2024, NeurIPS 2025, ICLR 2026, Neurocomputing journal